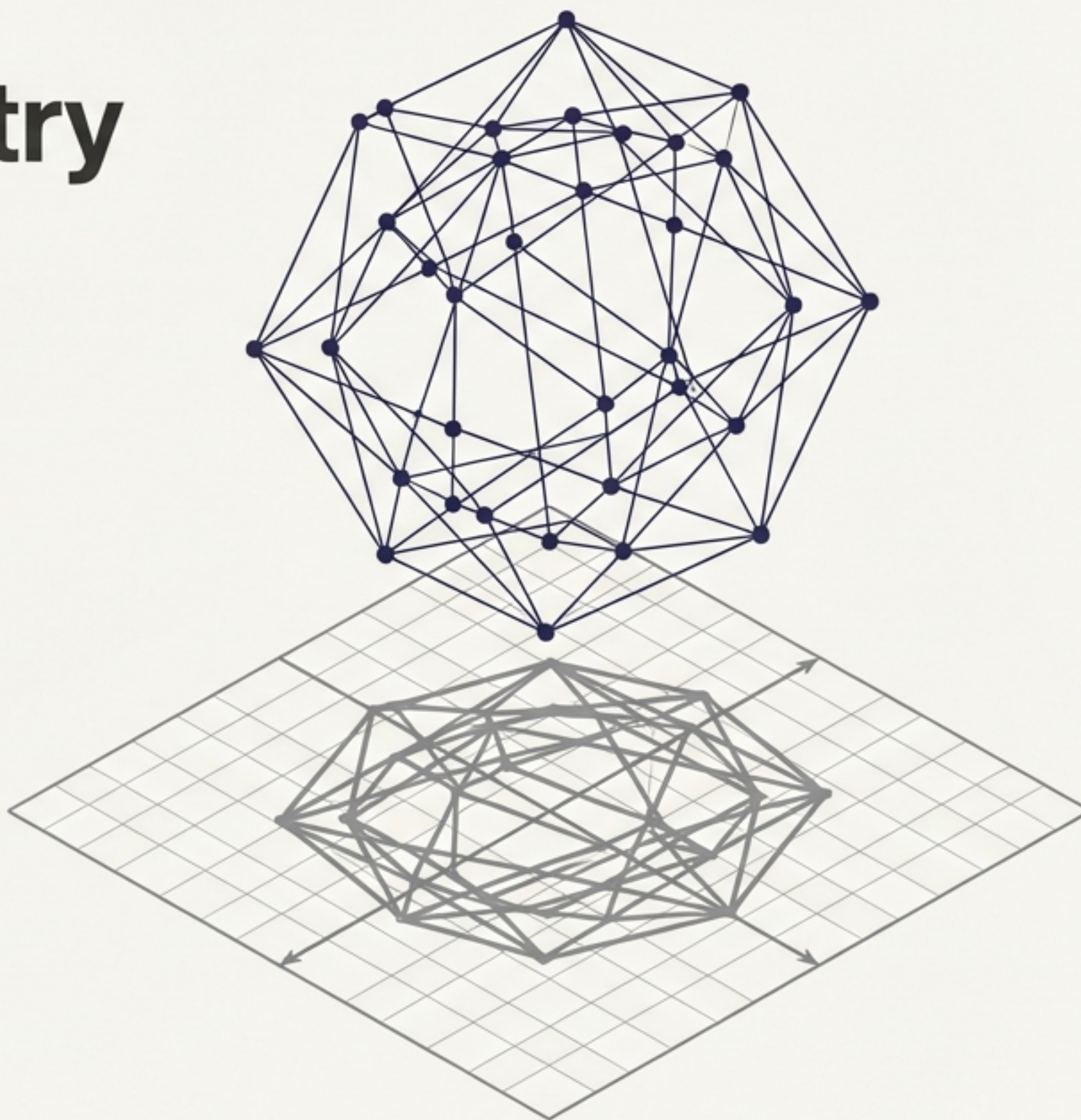


Principal Component Analysis: The Geometry of Information

From Geometric Intuition to Mathematical Formulation

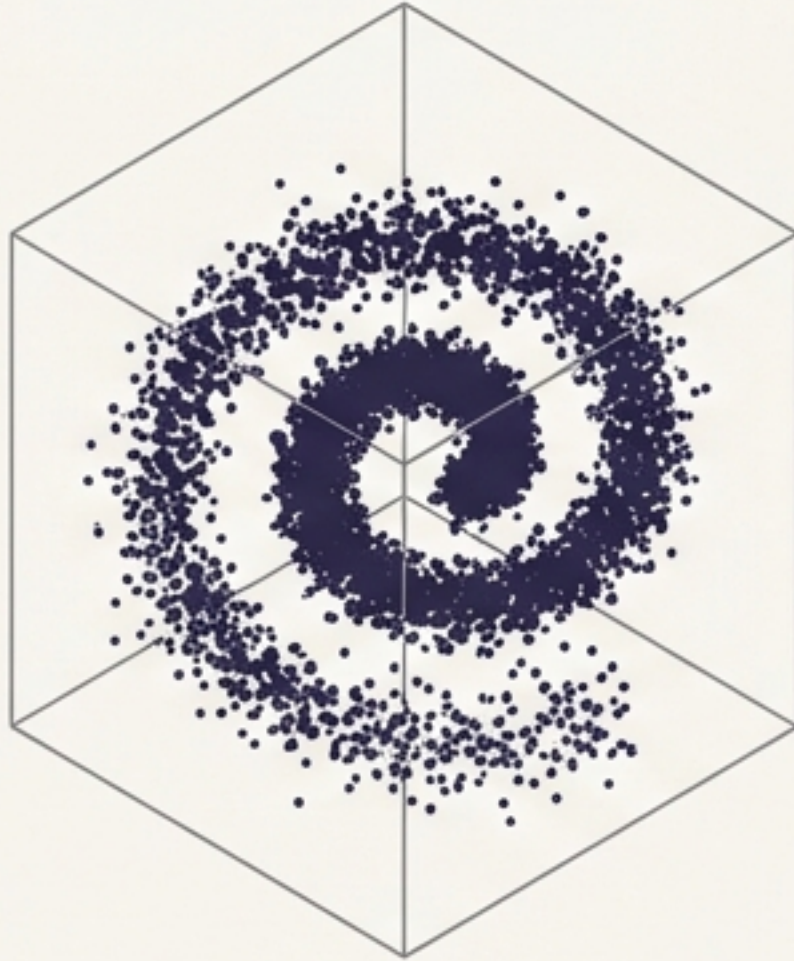
Principal Component Analysis (PCA) is often treated as a 'black box' for dimensionality reduction. This deck deconstructs the algorithm to reveal its core mathematical objective: finding the 'best shadow' of a high-dimensional dataset that preserves its essential variance.



Reducing Dimensions, Preserving Essence

The Goal of Dimensionality Reduction

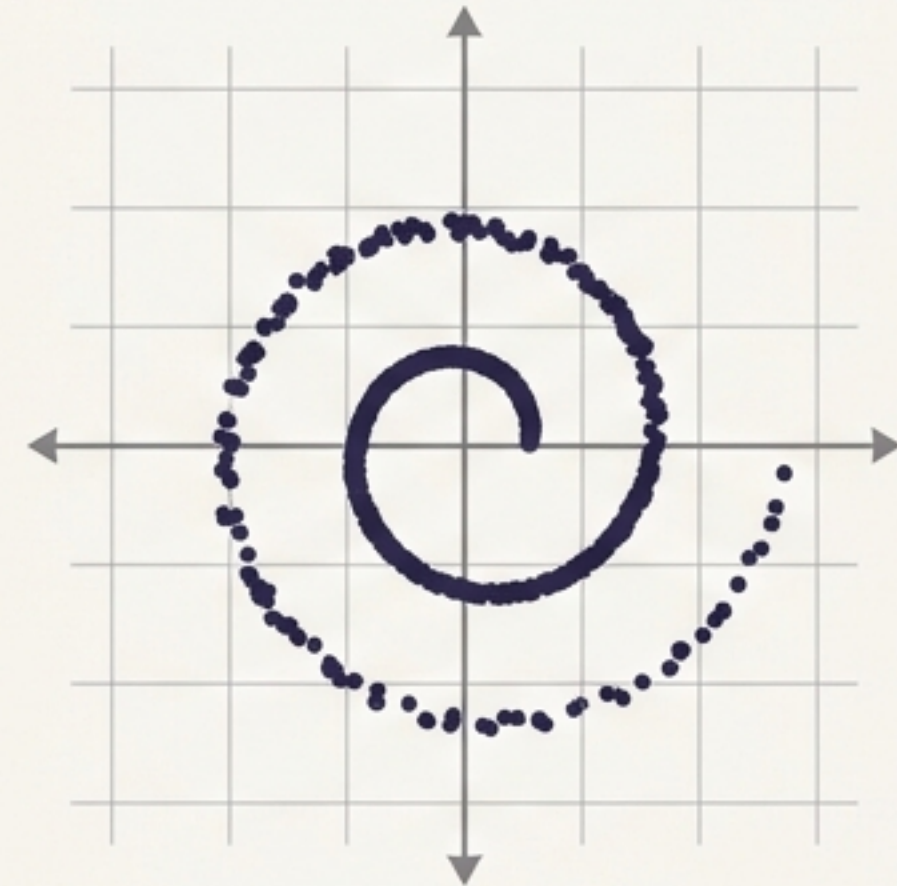
Roboto Mono
High-Dimensional Input (e.g., 3D)



Roboto Mono
Transformation



Roboto Mono
Lower-Dimensional Output (e.g., 2D)



The Goal

Reduce a dataset from high dimensions (many features) to lower dimensions (fewer features) to simplify processing.

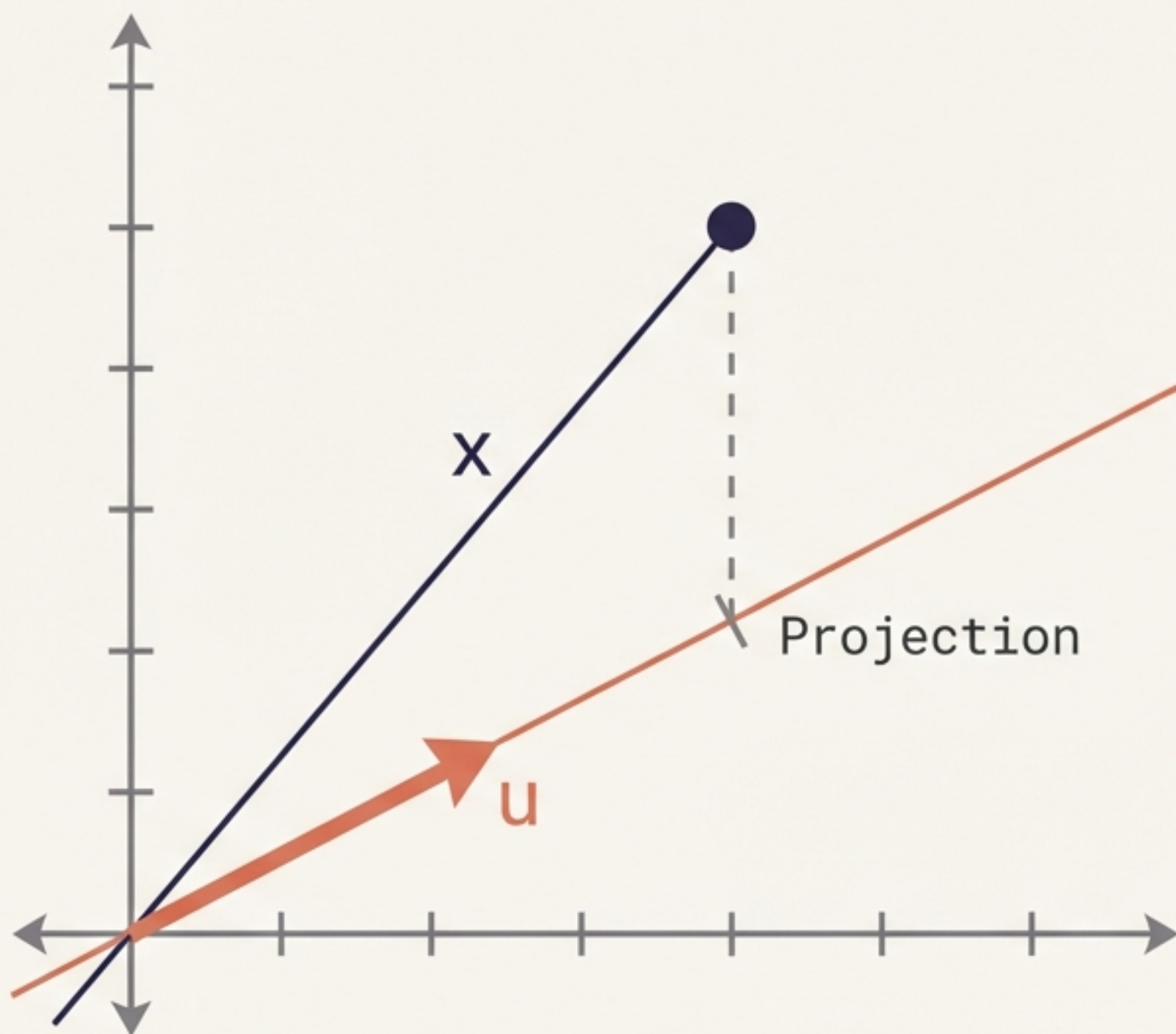
The Challenge

Simplifying the data without losing the unique signal that defines it. We must distinguish signal from noise.

The Approach

Find a projection that maintains the relationships and spread of the original data, ensuring machine learning models perform equally well on the reduced set.

The Geometric Problem: Finding the Right Axis.



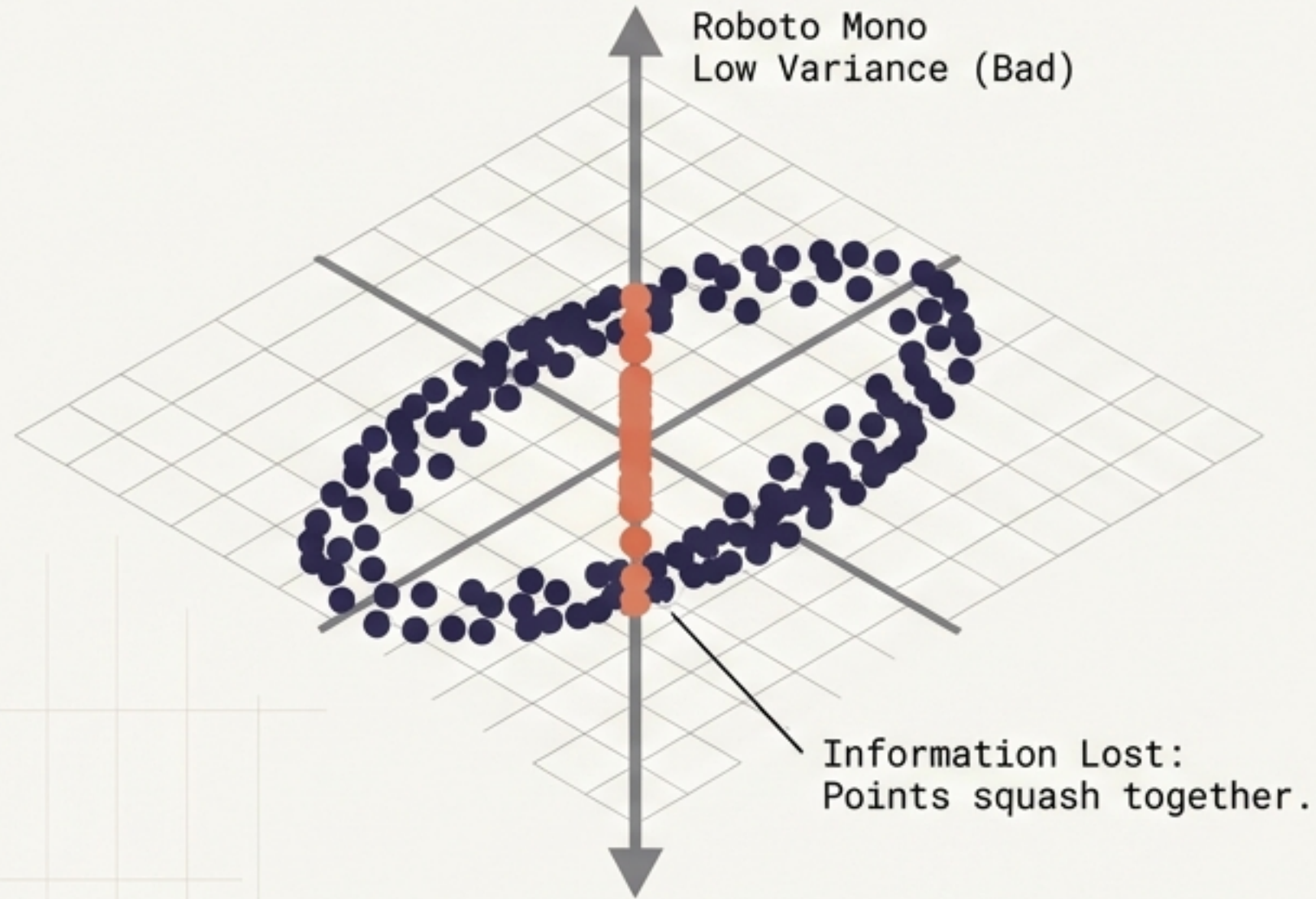
Imagine projecting 2D data onto a single line (1D). We treat every data point as a vector (x). We are searching for a specific direction, defined by a Unit Vector (u).

The mathematical operation is the projection of vector x onto unit vector u .

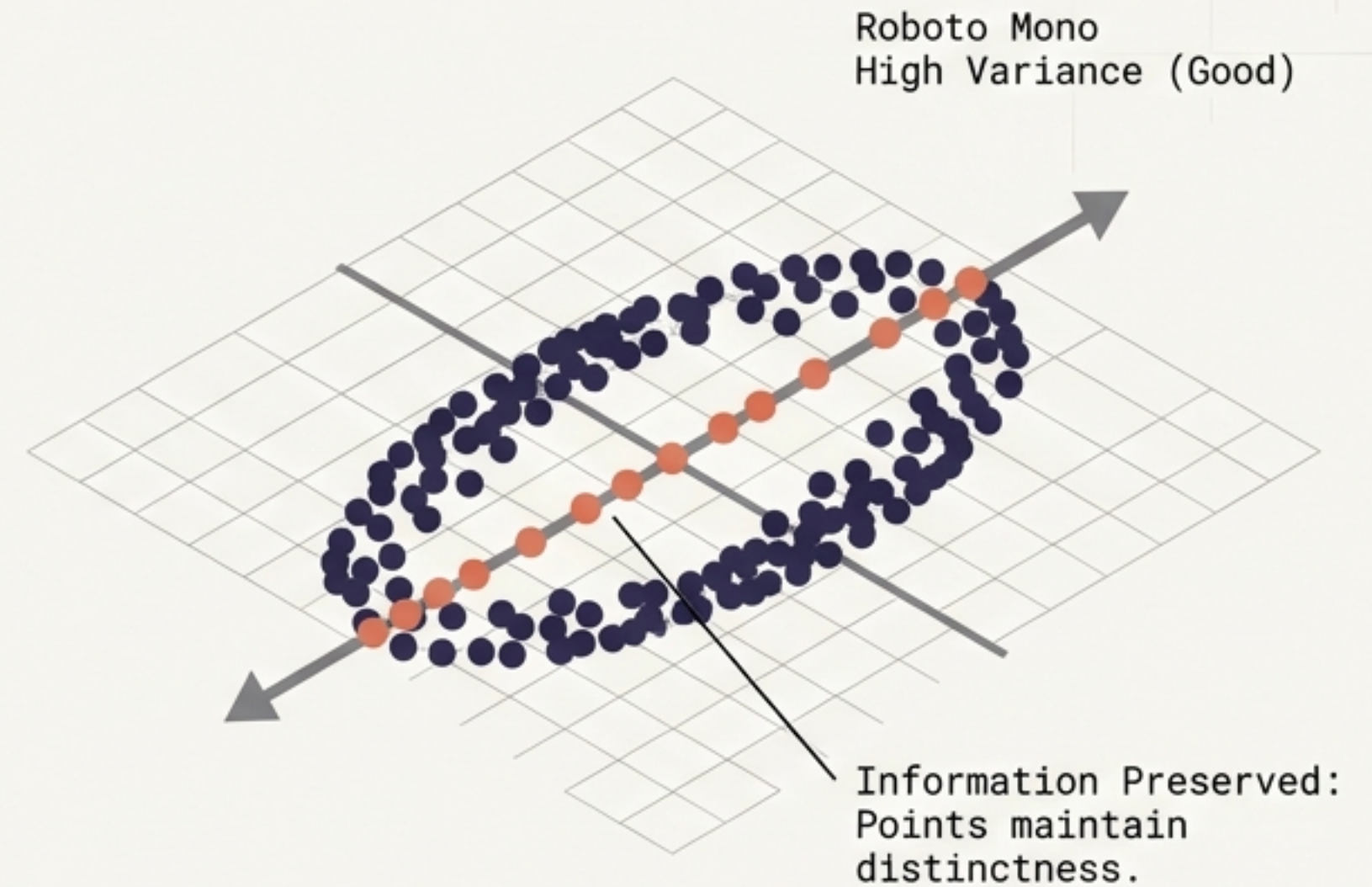
The Question: There are infinite directions for u (infinite ways to rotate the line). Which one creates the most useful shadow?

The Metric is Variance.

Scenario A - Bad



Scenario B - Good



We define the 'best' projection as the one that maximises the spread (variance) of the data.
If points are squashed together, we lose the ability to distinguish between them.
We must find the Unit Vector (u) where the projected points have the maximum possible spread.

The Mathematical Objective Function.

Maximise: $\frac{1}{n} \sum_{i=1}^n (\mathbf{u}^T \mathbf{x}_i)^2$

Average.

The Dot Product.
This calculates the length of the projection of vector $\mathbf{x}$ onto unit vector $\mathbf{u}$.

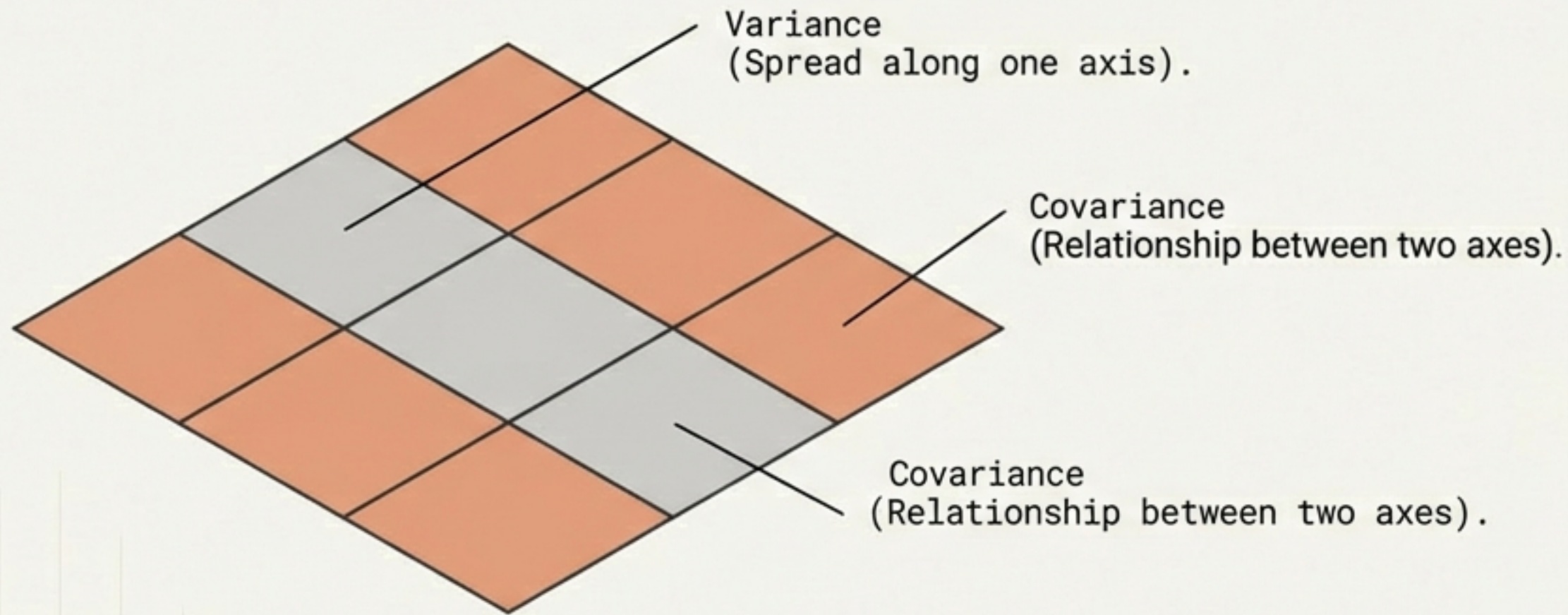
Sum over all data points.

The Variance of the projected data (assuming centred data where mean = 0).

To turn geometry into maths, we use the dot product. If we centre the data (subtracting the mean so $\bar{x} = 0$), the variance formula simplifies to the average of the squared projections.

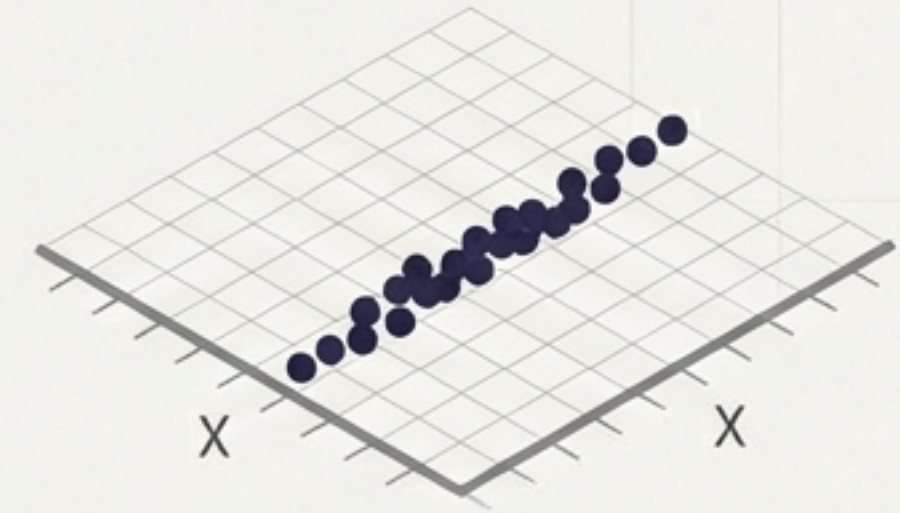
The Optimisation Problem: Find the vector $\mathbf{u}$ that yields the largest value for this equation. To solve this, we need two tools: The Covariance Matrix and Eigenvectors.

Tool 1: The Covariance Matrix.

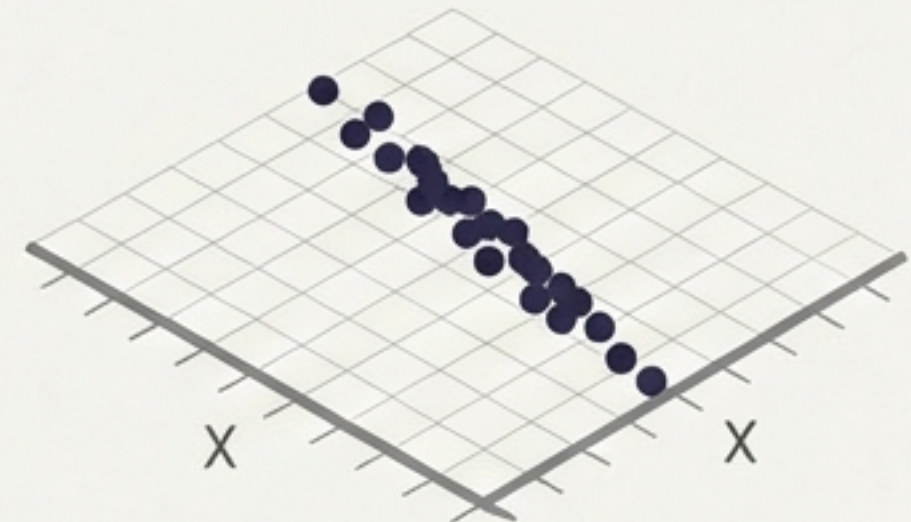


Variance measures spread along a single axis. Covariance measures the relationship *between* axes.

The Covariance Matrix is a symmetric grid that summarises the entire dataset. It captures not just how wide the data is spread, but its **orientation** in space.

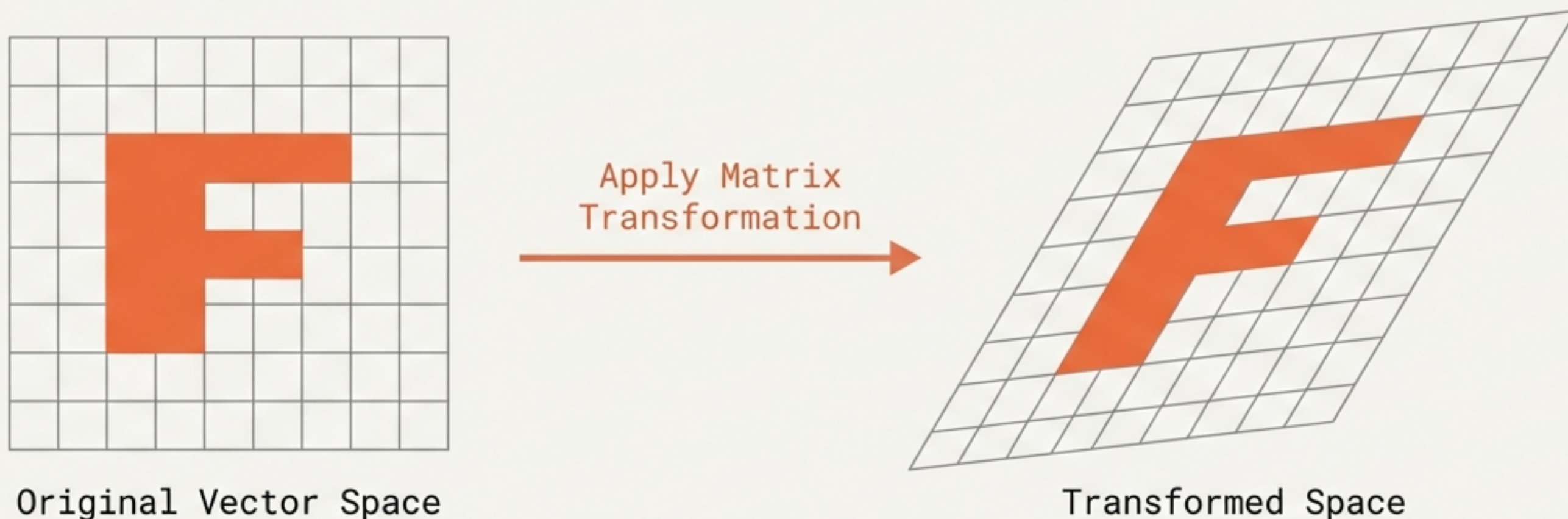


Positive Covariance
(As X increases, Y increases).



Negative Covariance
(As X increases, Y decreases).

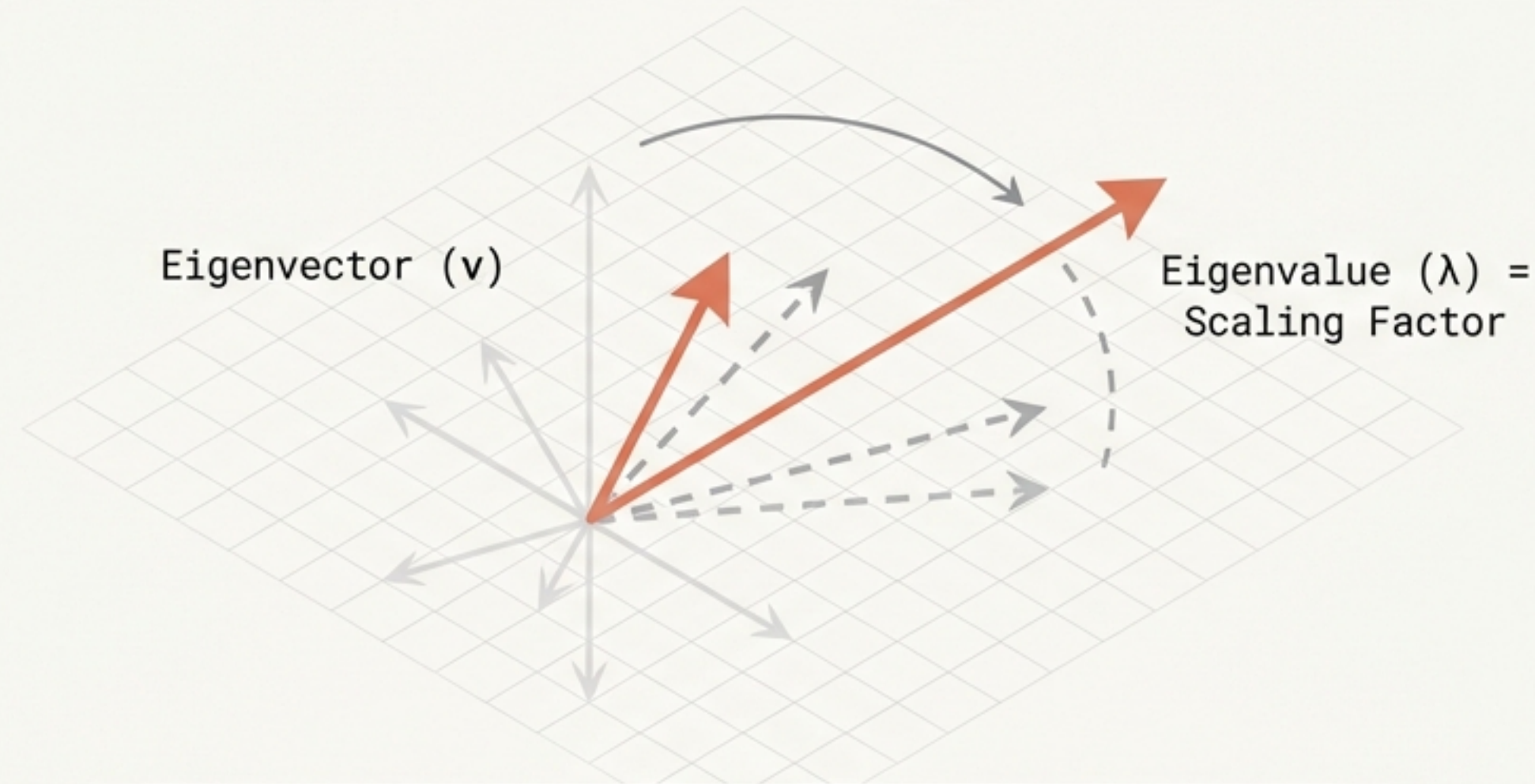
Tool 2: Matrices as Transformers



In linear algebra, a matrix is not just a table of numbers; it is a **transformation machine**.

When we apply a matrix to a vector space, it moves the points. It can rotate the space, stretch it, or shear it. This concept is crucial because PCA involves finding vectors that behave “predictably” under the transformation defined by the Covariance Matrix.

Tool 3: Eigenvectors & Eigenvalues

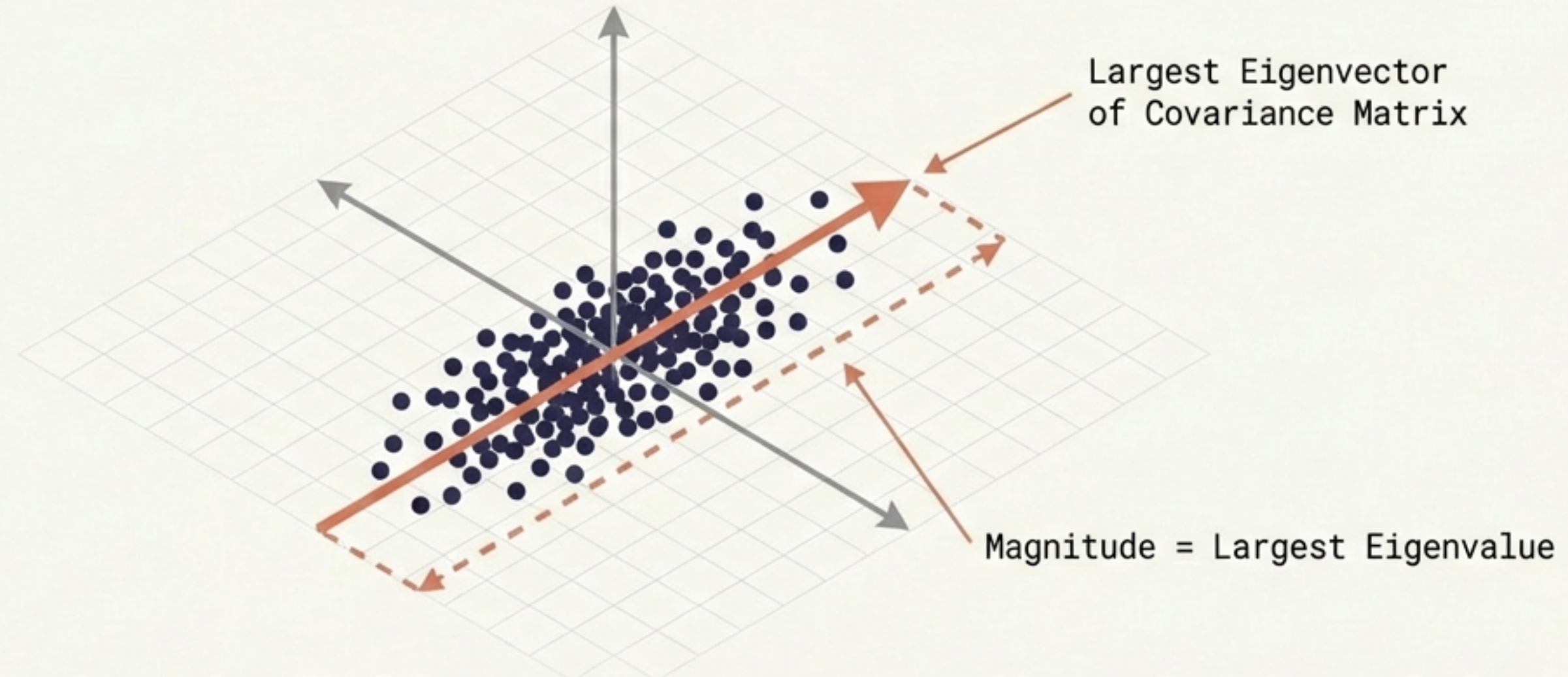


$$A \mathbf{v} = \lambda \mathbf{v}$$

Matrix (A) × Vector (v) = Scalar (λ) × Vector (v)

During a linear transformation, most vectors get knocked off their original span (direction). **Eigenvectors** are special: they do not change direction; they are only scaled (stretched or shrunk). The **Eigenvalue** tells us how much stretching occurred.

The Solution: The 'Aha!' Moment.

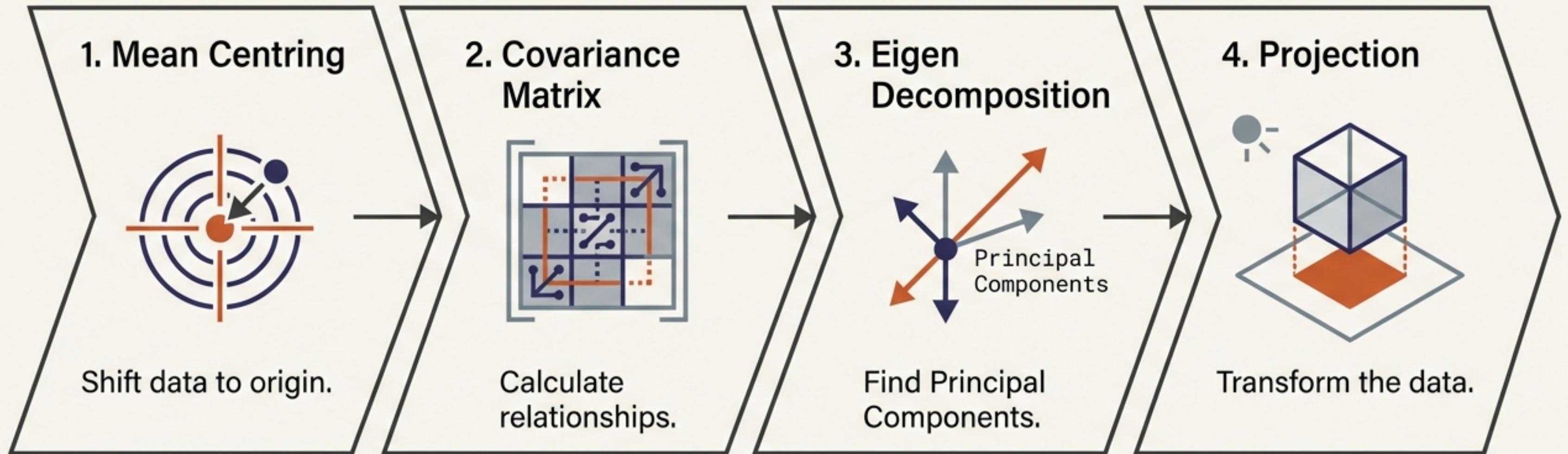


Mathematicians discovered a profound link between our optimisation problem and our linear algebra tools.

The Core Truth: The direction of maximum variance (our 'best shadow') is exactly the **Eigenvector** of the data's CovCovariance Matrix that possesses the largest **Eigenvalue**.

We do not need to guess axes; we simply decompose the Covariance Matrix to find the 'natural axes' of the data.

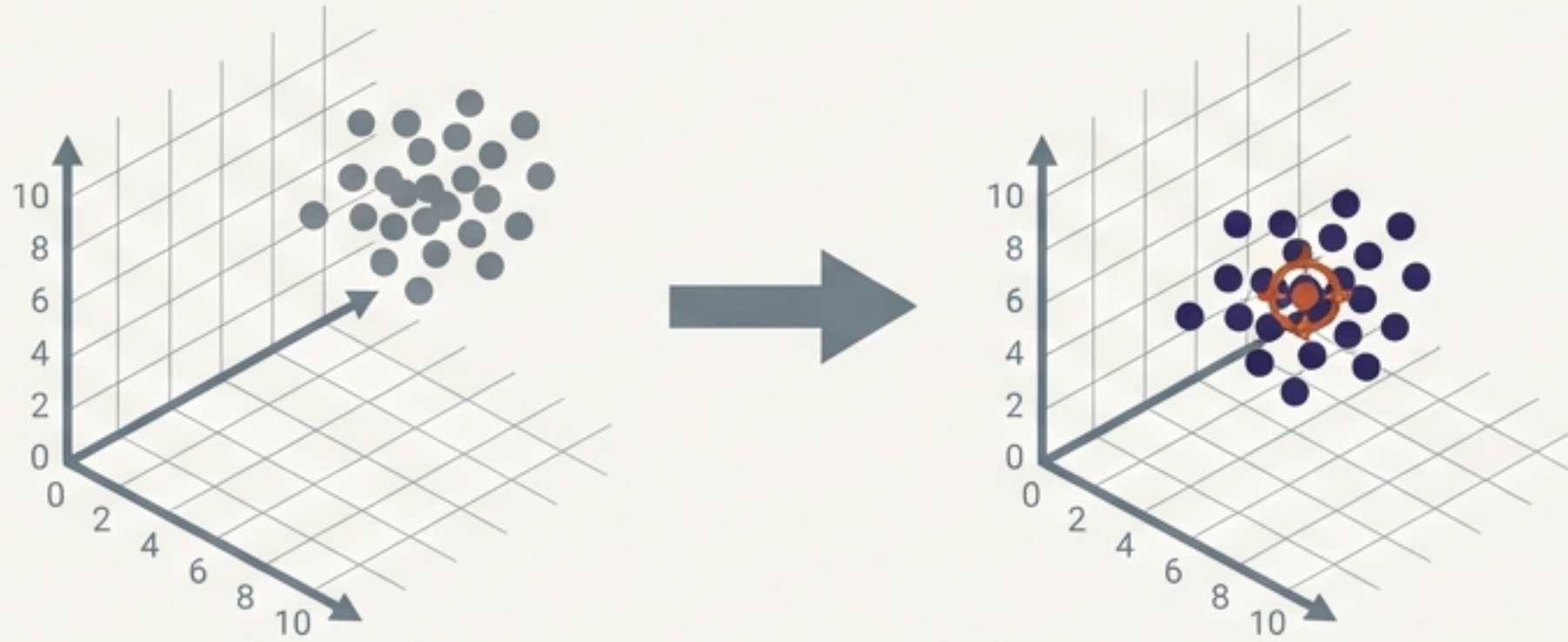
The Algorithm: Step-by-Step.



We now have a repeatable recipe to transform high-dimensional data into its principal components.

Step 1 & 2: Centring and Covariance.

Centring



Subtract mean from every data point. $\bar{x} \rightarrow 0$.

Covariance

1.5	0.4	0.2
0.4	2.0	0.1
0.2	0.1	0.8

Compute Matrix on centred data.
This captures the spread and
orientation in N-dimensional space.

Step 1: Mean Centring. Moving the dataset to the origin simplifies the maths.

Step 2: Covariance Matrix. We calculate how every feature relates to every other feature.

Step 3: Finding Principal Components

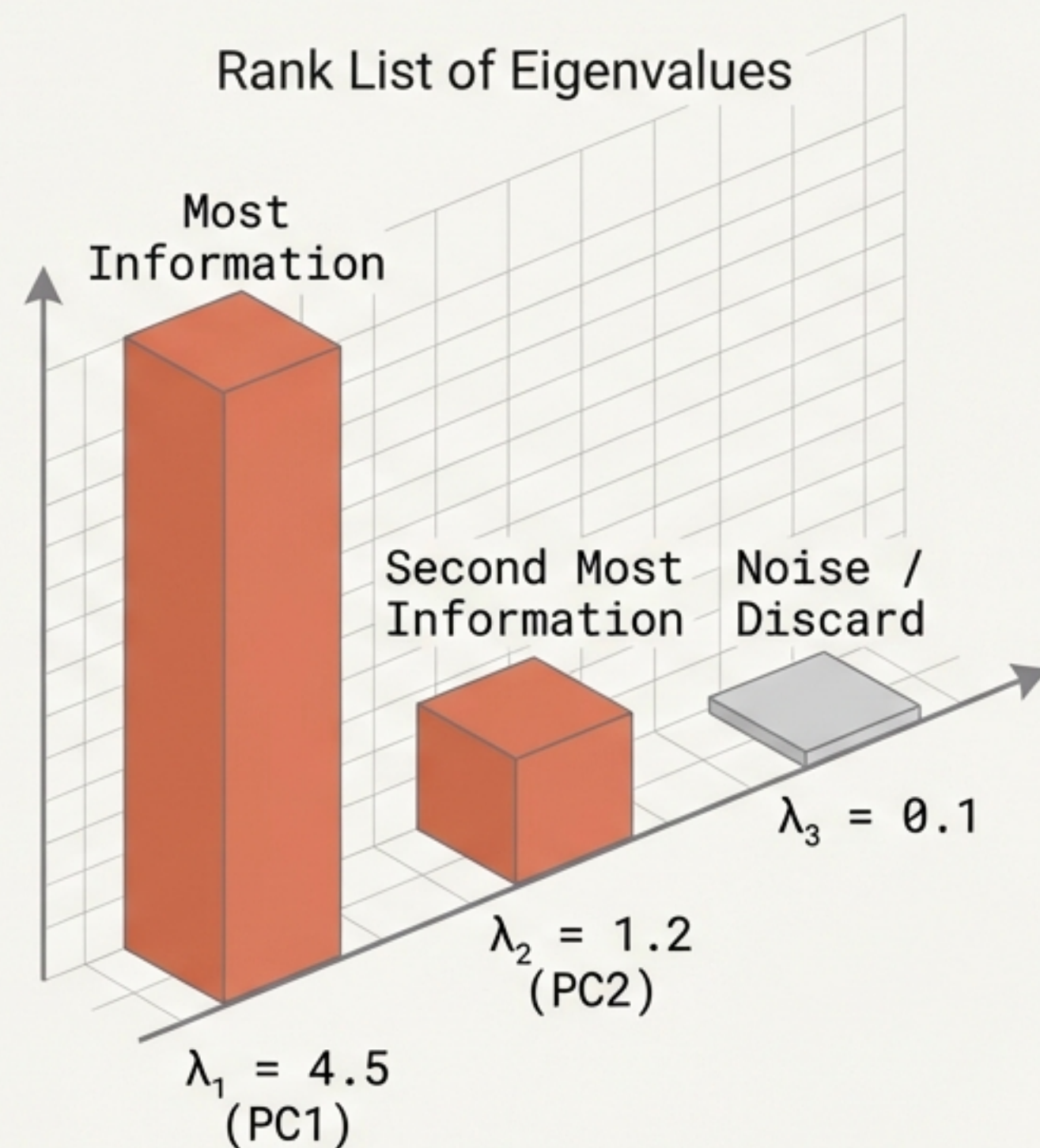
Calculate Eigenvectors and Eigenvalues of the Covariance Matrix.

Sorting: Rank vectors from high λ to low λ .

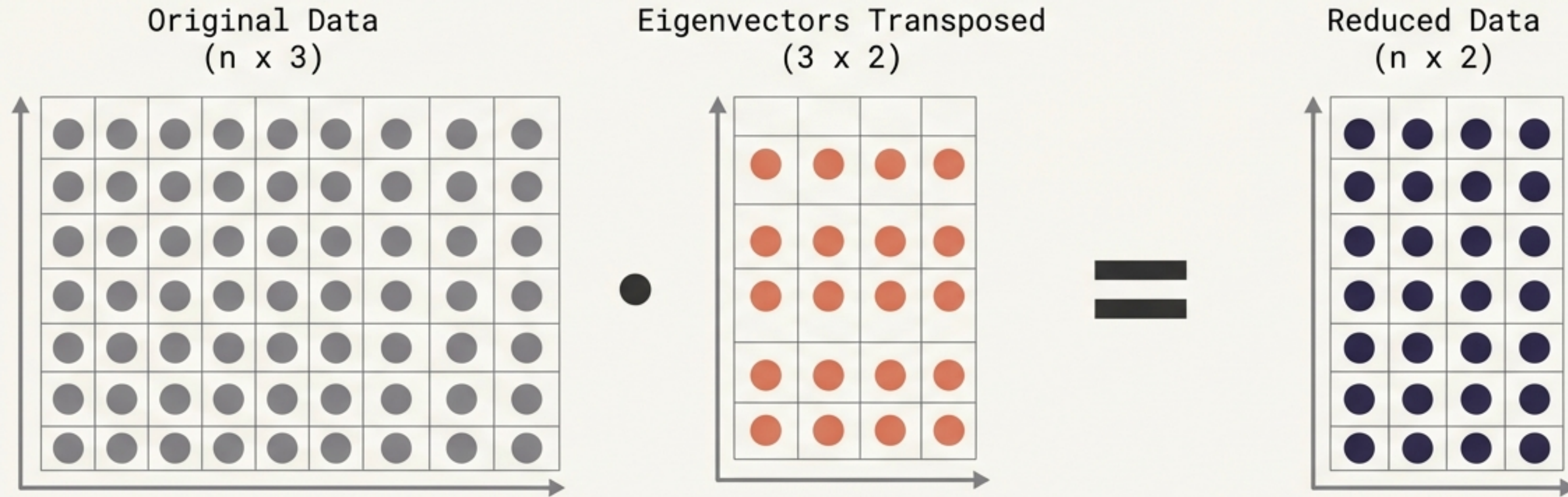
Selection: Pick the top k vectors.

These are your Principal Components.

PC1 is the vector with the largest Eigenvalue.



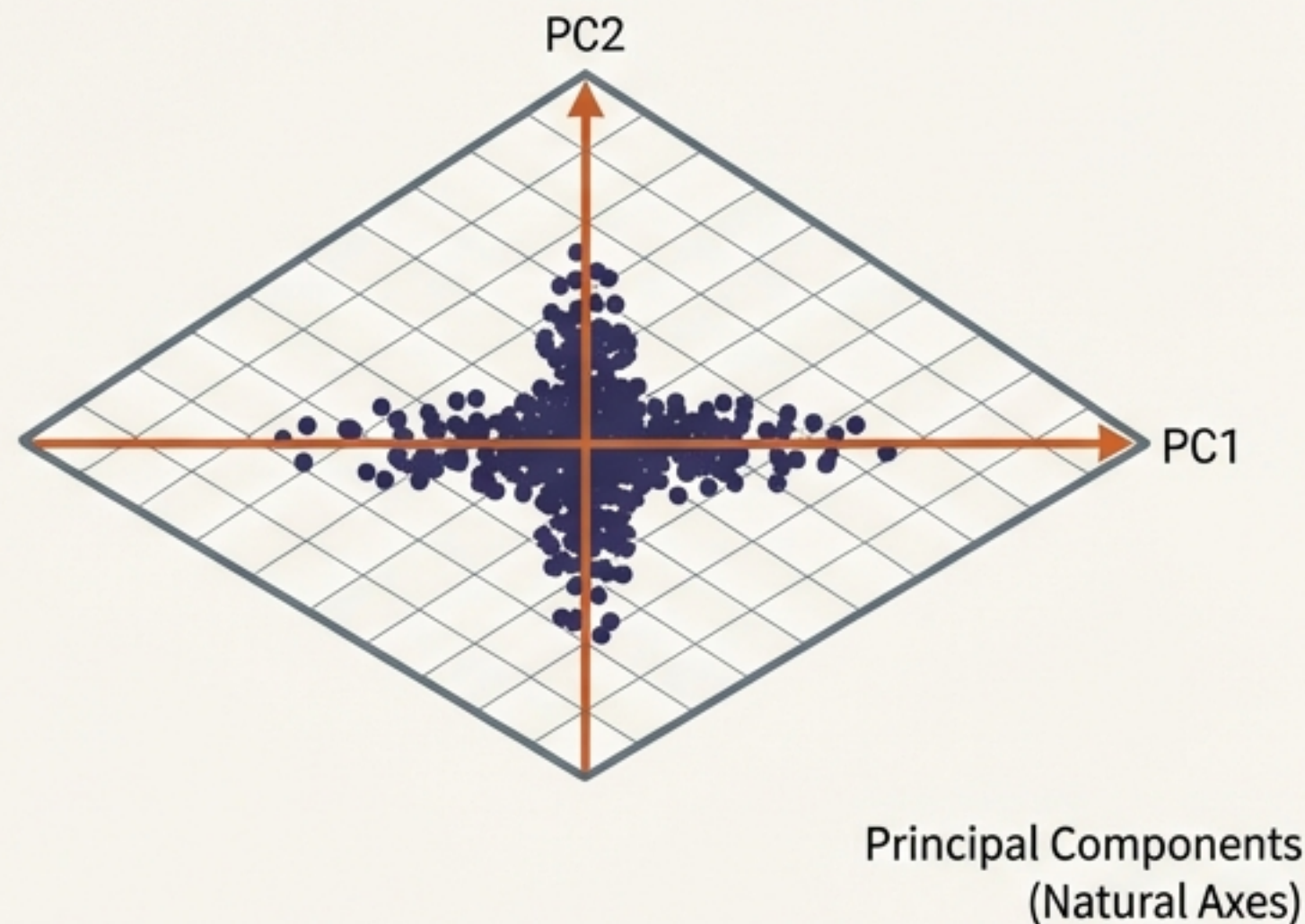
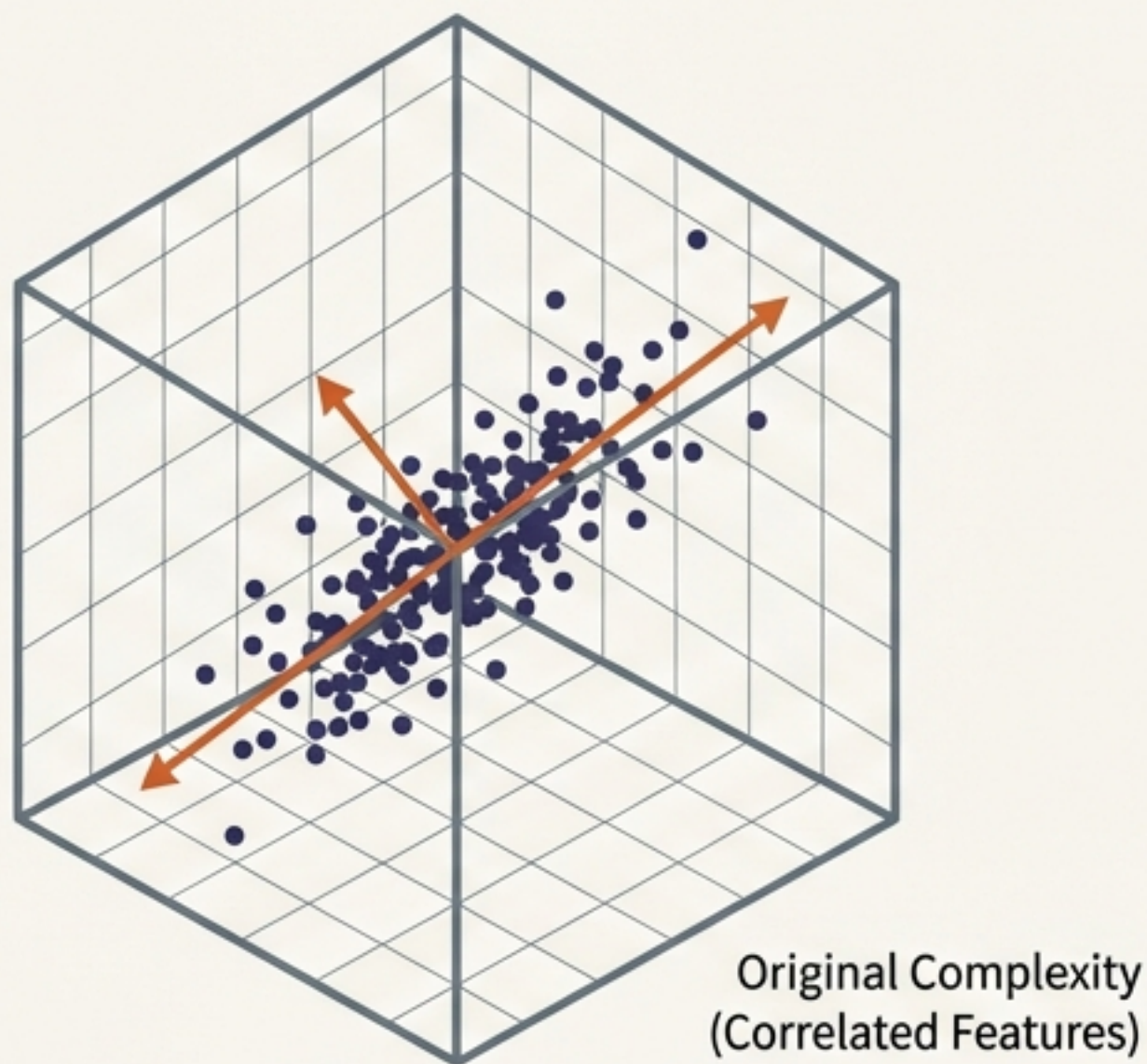
Step 4: The Projection.



Finally, we project the original data onto the selected Principal Components.

- **The Operation:** Dot product of Original Data and the Transpose of selected Eigenvectors.
- **The Result:** A new dataset where columns are no longer 'Feature 1, Feature 2' but 'PC1, PC2'. We have successfully reduced dimensions whilst retaining maximum variance.

Summary: Finding the Natural Axes



- **Intuition:** PCA rotates the coordinate system to align with the data's natural spread.
- **Maths:** It solves an optimisation problem (max variance) using Linear Algebra (Eigenvectors).
- **Result:** We simplify complexity. We reduce noise and dimensions while preserving the signal—the 'essence' of the data.